

## 78 independent Lorentz scalars

Created by David Rabinow

Autonomous numerical discovery · chiral 5-form in ten dimensions

Numerically independent invariants after product filtering. Literature target  $\approx 81$  primary generators.

Catalog entries use ordinary math notation — e.g.  $\text{tr}(T^2)$ ,  $\|G\|^2$ ,  $\text{tr}(T^3)$ . Graph entries are written  $G_4$ ,  $G_6$ ,  $G_8$ , ... for automatic contraction patterns (subscript = number of tensor copies).

Breakdown: 6 catalog · 72 graphs | orders: 3 at order 4, 6 at order 6, 66 at order 8, 3 at order 10

### Order 4 (3 invariants)

01.  $\text{tr}(T^2)$   
catalog · degree 4
02.  $\|G\|^2$   
catalog · degree 4
03.  $G_4 \text{ M}[1,2,2,2,2,1]$  (alt, pop)  
graph · degree 4

### Order 6 (6 invariants)

04.  $\text{tr}(T^3)$   
catalog · degree 6
05.  $G_6 \text{ M}[0,0,0,1,4,0,1,3,\dots]$  (alt, pop)  
graph · degree 6
06.  $G_6 \text{ M}[0,0,0,1,4,0,1,3,\dots]$  (half, pop)  
graph · degree 6
07.  $G_6 \text{ M}[0,0,0,1,4,1,1,2,\dots]$  (first, pop)  
graph · degree 6
08.  $G_6 \text{ M}[0,0,1,2,2,1,1,1,\dots]$  (alt, pop)  
graph · degree 6
09.  $G_6 \text{ M}[0,0,1,2,2,2,1,0,\dots]$  (alt, pop)  
graph · degree 6

### Order 8 (66 invariants)

10.  $\text{tr}(T^4)$   
catalog · degree 8
11.  $\text{tr}(M_G^2)$   
catalog · degree 8
12.  $G_8 \text{ WL}[(5, 5, 5, 5, 5, 5, 5, 5)]((0, 1), (0, 2), (0, 3)\dots]$  (alt, pop)  
graph · degree 8
13.  $G_8 \text{ WL}[(5, 5, 5, 5, 5, 5, 5, 5)]((0, 1), (0, 2), (0, 3)\dots]$  (half, pop)  
graph · degree 8
14.  $G_8 \text{ WL}[(5, 5, 5, 5, 5, 5, 5, 5)]((0, 1), (0, 2), (0, 3)\dots]$  (first, pop)  
graph · degree 8
15.  $G_8 \text{ WL}[(5, 5, 5, 5, 5, 5, 5, 5)]((0, 1), (0, 2), (0, 5)\dots]$  (alt, pop)  
graph · degree 8
16.  $G_8 \text{ WL}[(5, 5, 5, 5, 5, 5, 5, 5)]((0, 1), (0, 2), (0, 3)\dots]$  (alt, pop)  
graph · degree 8

17.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 3), (0, 4)...)] (alt, pop)  
graph · degree 8
18.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 4)...)] (alt, pop)  
graph · degree 8
19.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 4)...)] (half, pop)  
graph · degree 8
20.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 4)...)] (first, pop)  
graph · degree 8
21.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 2)...)] (alt, pop)  
graph · degree 8
22.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 2), (0, 4)...)] (alt, pop)  
graph · degree 8
23.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 2), (0, 4)...)] (half, pop)  
graph · degree 8
24.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 2), (0, 4)...)] (first, pop)  
graph · degree 8
25.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 2)...)] (alt, pop)  
graph · degree 8
26.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 4), (0, 6)...)] (alt, pop)  
graph · degree 8
27.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 4), (0, 6)...)] (half, pop)  
graph · degree 8
28.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 4), (0, 6)...)] (first, pop)  
graph · degree 8
29.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 4)...)] (alt, pop)  
graph · degree 8
30.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 5), (0, 5), (0, 6)...)] (alt, pop)  
graph · degree 8
31.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 5), (0, 5), (0, 6)...)] (half, pop)  
graph · degree 8
32.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 5), (0, 5), (0, 6)...)] (first, pop)  
graph · degree 8
33.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 3)...)] (alt, pop)  
graph · degree 8
34.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 3)...)] (alt, pop)  
graph · degree 8
35.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 4), (0, 4)...)] (alt, pop)  
graph · degree 8
36.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 3), (0, 4), (0, 4)...)] (alt, pop)  
graph · degree 8
37.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 3), (0, 4), (0, 4)...)] (half, pop)  
graph · degree 8
38.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 3), (0, 4), (0, 4)...)] (first, pop)  
graph · degree 8
39.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 4), (0, 4)...)] (alt, pop)  
graph · degree 8
40.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 4), (0, 5)...)] (alt, pop)  
graph · degree 8

41.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 4), (0, 5)...)] (half, pop)  
graph · degree 8
42.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 4), (0, 5)...)] (first, pop)  
graph · degree 8
43.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 4)...)] (alt, pop)  
graph · degree 8
44.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 4)...)] (alt, pop)  
graph · degree 8
45.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 4)...)] (half, pop)  
graph · degree 8
46.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 4)...)] (first, pop)  
graph · degree 8
47.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 3), (0, 3)...)] (alt, pop)  
graph · degree 8
48.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 3), (0, 6)...)] (alt, pop)  
graph · degree 8
49.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 3), (0, 5)...)] (alt, pop)  
graph · degree 8
50.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 3), (0, 5)...)] (half, pop)  
graph · degree 8
51.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 3), (0, 5)...)] (first, pop)  
graph · degree 8
52.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 3)...)] (alt, pop)  
graph · degree 8
53.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 3)...)] (half, pop)  
graph · degree 8
54.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 3), (0, 4)...)] (alt, pop)  
graph · degree 8
55.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 4)...)] (alt, pop)  
graph · degree 8
56.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 4)...)] (half, pop)  
graph · degree 8
57.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 4)...)] (first, pop)  
graph · degree 8
58.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 5), (0, 5)...)] (alt, pop)  
graph · degree 8
59.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 2)...)] (alt, pop)  
graph · degree 8
60.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 2)...)] (half, pop)  
graph · degree 8
61.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 2)...)] (first, pop)  
graph · degree 8
62.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 1), (0, 3)...)] (alt, pop)  
graph · degree 8
63.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 4)...)] (alt, pop)  
graph · degree 8
64.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 6), (0, 6)...)] (alt, pop)  
graph · degree 8

65.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 6), (0, 6)...)] (half, pop)  
graph · degree 8
66.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 6), (0, 6)...)] (first, pop)  
graph · degree 8
67.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 5), (0, 5)...)] (alt, pop)  
graph · degree 8
68.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 4), (0, 4)...)] (alt, pop)  
graph · degree 8
69.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 4), (0, 4)...)] (half, pop)  
graph · degree 8
70.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 4), (0, 4)...)] (first, pop)  
graph · degree 8
71.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 3), (0, 5)...)] (half, pop)  
graph · degree 8
72.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 3), (0, 3)...)] (alt, pop)  
graph · degree 8
73.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 3), (0, 3)...)] (half, pop)  
graph · degree 8
74.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 2), (0, 3), (0, 3)...)] (first, pop)  
graph · degree 8
75.  $G_8$  WL[(5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 2), (0, 2)...)] (alt, pop)  
graph · degree 8

## Order 10 (3 invariants)

76.  $\text{tr}(T^5)$   
catalog · degree 10
77.  $G_{10}$  WL[(5, 5, 5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 3), ...)] (alt, pop)  
graph · degree 10
78.  $G_{10}$  WL[(5, 5, 5, 5, 5, 5, 5, 5, 5, 5)|((0, 1), (0, 3), ...)] (alt, pop)  
graph · degree 10

Superscripts denote powers ( $T^2$  means T squared), not footnotes. Graph IDs are shortened fingerprints of the contraction pattern. alt / half / first = Lorentz raise patterns; pop / pop0 = index-wiring policies.